

Take a Demo Test

verify your email to take a free demo test

Select Exam

Select Exam

Your Name

Enter Your Name

Email Address

Enter your Email Address

Verify Email

Enter 6 digit otp

Enter OTP

Verify OTP

Otp has been sent to [@example.email.com](#)
Please verify it

Submit

Please read the instructions

- 1.Total duration of the examination is 180 minutes.
- 2.The clock will be set at the server. The countdown timer in the top right corner of screen will display the remaining time available for you to complete the examination. When the timer reaches zero, the examination will end by itself. You will not be required to end or submit your examination.
- 3.The Question Palette displayed on the right side of screen will show the status of each question using one of the following symbols:

- 1

You have NOT visited the question yet.
- 2

You have NOT answered the question.
- 3

You have answered the question.
- 4

You have marked the question for review.

You can click on the arrow which appears to the left of question palette to collapse the question palette thereby maximizing the question window. To view the question palette again, you can click on which appears on the right side of question window.

Paper Specific Instructions

This question paper has 65 questions for a total of 100 marks. It consists of two sections: General Aptitude (GA) for 15 marks and the subject specific section for 85 marks. Both sections are compulsory. The marks distribution is as follows.

Section	No. of 1 mark questions	No of 2 mark questions
General Aptitude	5	5
Subject Specific	25	30

☐

I have read and understood the instructions

Proceed with
the Test

Subjects:

Maths

Physics

Chemistry

Sections:

Section 1

Section 2

Question Palette

1

Not Visited

3

Unanswered

2

Answered

4

Marked for review

1

2

3

4

5

6

7

8

9

10

Question no : 1

Marks +4/-1

Question Image

Option 1

Option 2

Option 3

Option 4

Mark for review

Previous

Save and
Next

Submit Test

Test results

Total Marks:

280

out of 300

Correct Questions:

50

out of 90

Wrong Questions:

20

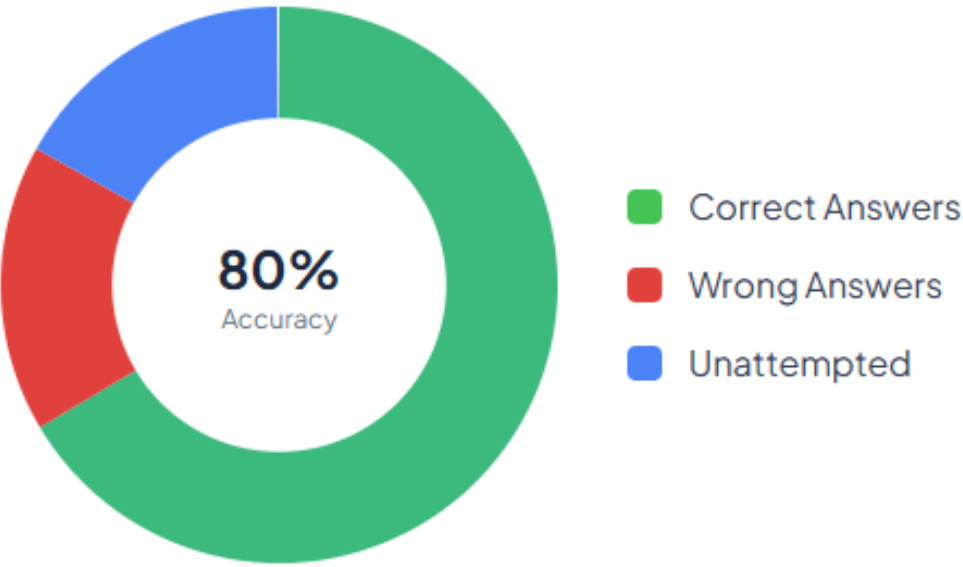
out of 90

Un Attempted:

20

out of 90

Overall Test Performance



50*4=200

20*-1=-20

20*0=0

Subject-wise Performance

Physics

76/100

Marks Obtained

19 Correct

4 Wrong

7 Skipped

View Solutions >

Chemistry

84/100

Marks Obtained

21 Correct

5 Wrong

4 Skipped

View Solutions >

Mathematics

80/100

Marks Obtained

20 Correct

6 Wrong

4 Skipped

View Solutions >

Questions

- 1
- 2
- 3
- 4
- 5
- 6
- 7
- 8
- 9
- 10
- 11
- 12
- 13

Let $a_1, a_2, a_3, \dots$ be an arithmetic progression with $a_1 = 7$ and common difference 8. Let $T_1, T_2, T_3, \dots$ be such that $T_1 = 3$ and $T_{n+1} - T_n = a_n$ for $n \geq 1$. Then, which of the following is/are TRUE ?

- (A) $T_{20} = 1604$
- (B) $\sum_{k=1}^{20} T_k = 10510$
- (C) $T_{30} = 3454$
- (D) $\sum_{k=1}^{30} T_k = 35610$

Your Answer

Correct Answer

Sol. Here $a_n = 7 + (n - 1) 8$ and $T_1 = 3$

Also $T_{n+1} = T_n + a_n$

$$T_n = T_{n-1} + a_{n-1}$$

$\vdots$

$$T_2 = T_1 + a_1$$

$$\therefore T_{n+1} = (T_{n-1} + a_{n-1}) + a_n$$

$$= T_{n-2} + a_{n-2} + a_{n-1} + a_n$$

$\vdots$

$$\Rightarrow T_{n+1} = T_1 + a_1 + a_2 + \dots + a_n$$

$$\Rightarrow T_{n+1} = T_1 + \frac{n}{2} [2(7) + (n-1)8]$$

$$\Rightarrow T_{n+1} = T_1 + n(4n+3) \quad \dots (1)$$

$$\therefore \text{For } n = 19 \quad T_{20} = 3 + (19)(79) = 1504$$

$$\text{For } n = 29 \quad T_{30} = 3 + (29)(119) = 3454 \rightarrow (C)$$

$$\sum_{k=1}^{20} T_k = 3 + \sum_{k=2}^{20} T_k = 3 + \sum_{k=1}^{19} (3 + 4n^2 + 3n)$$

$$= 3 + 3(19) + \frac{3(19)(20)}{2} + \frac{4(19)(20)(39)}{6}$$

$$= 3 + 10507 = 10510 \rightarrow (B)$$

$$\text{And Similarly } \sum_{k=1}^{30} T_k = 3 + \sum_{k=1}^{29} (4n^2 + 3n + 3) = 35615$$

Vedio Solution

